

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460785.741 +/- 0.013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.104 +/- 0.022
Transit depth (Rp/Rs)^2	1.08 +/- 0.45 [%]
Orbital Inclination (inc)	85.67 +/- 1.48
Airmass coefficient 1 (a1)	1.1652 +/- 0.0086
Airmass coefficient 2 (a2)	-0.101 +/- 0.016
Scatter in the residuals of the lightcurve fit is	0.73 %
Best Comparison Star	None
Optimal Aperture	3.47
Optimal Annulus	7.22
Transit Duration (day)	0.09 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.104 ± 0.022 vs 0.1276 (deviation 1.07σ)
Duration fit vs published	0.09 d vs 0.1179 d (deviation 0.9σ)
Mid-time O–C	7.2 min
Depth SNR	4.7
Residual scatter	0.73 %

Light curve

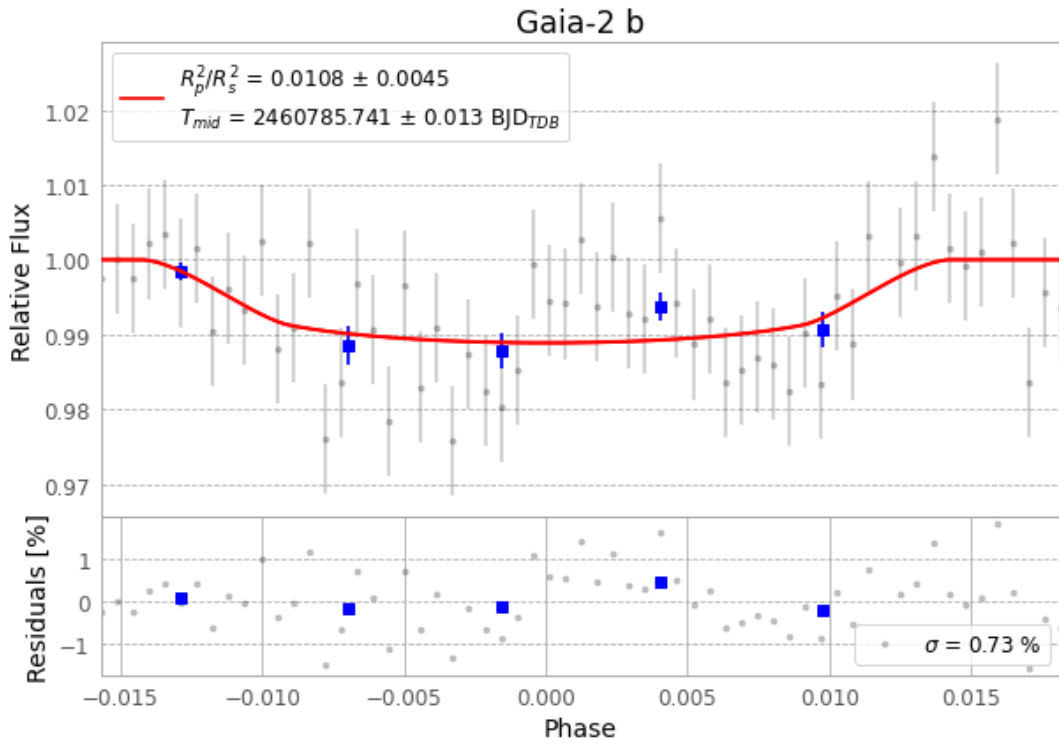


Figure 1 — FinalLightCurve_Gaia-2 b_2025-04-19.png (SHA-256 c187c27fd9c52c34...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [321, 302], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4d51a55ab93da251...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

61 FITS frames (40 MB), Gaia-2250420042416.FITS → Gaia-2250420072415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Gaia-2-250420.log (ca71c4f33efb...), FinalLightCurve_Gaia-2 b_2025-04-19.png (c187c27fd9c5...), FinalLightCurve_Gaia-2 b_2025-04-19.csv (8a82c3a5523f...)

Compute resources

Wall time 141.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 22:49Z.